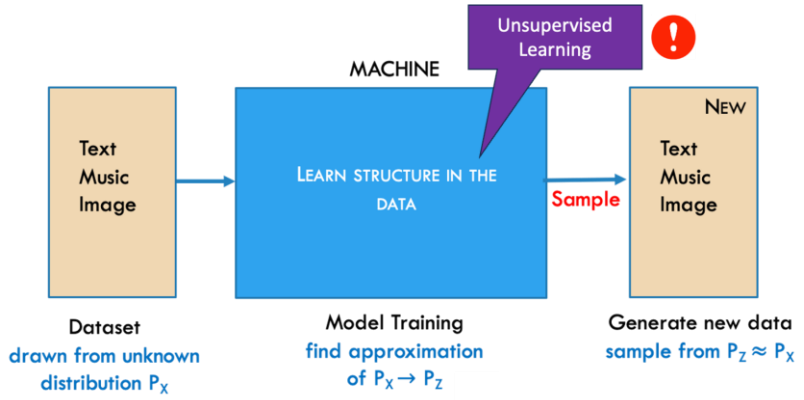
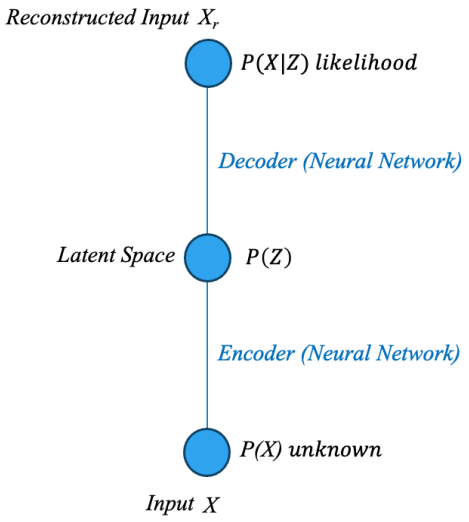
Reading 7.1: Mathematical Foundation of VAEs

This learning unit explains the mathematical foundation of Variational Autoencoders (VAEs). It covers defining distributions P(Z) and P(X|Z), and optimising the variational lower bound for log P(X).



We define two distributions P(Z) and P(X|Z), where P(X|Z) is the likelihood of seeing a reconstructed output (Xr) after sampling from Z. Let's start from the definition of the joint probability distribution P(X,Z) and Bayes Rule.



Joint Probability $P(X,Z) = P(X|Z).P(Z)$ (1)

(1) $\rightarrow$ $P(X|Z) = \frac{P(X,Z)}{P(Z)}$

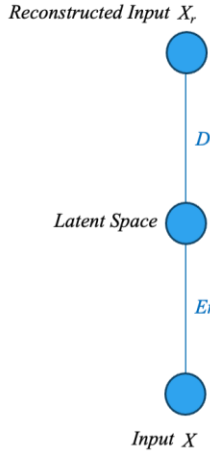
Bayes Rule $P(Z|X) = \frac{P(X|Z).P(Z)}{P(X)}$ (2) *Likelihood P(X|Z) is unknown*

Marginal Probability $P(X) = \int_{-\infty}^{+\infty} P(Z)\,P(X|Z)dZ$ *Integral is Intractable Z is high dimensional*

P(X) and the likelihood P(X|Z) are unknown and the integral to calculate P(X) is intractable for following reasons:

1. Z often represents a high-dimensional latent space and computing the integral requires summing over all possible Z values in that space, which becomes computationally expensive or impossible.
2. P(X|Z) may not have a closed-form solution
3. Even if P(Z) is a simple prior like a Gaussian, the interaction P(Z).P(X|Z) might result in a non-analytical form.

Instead of maximising the likelihood P(X|Z), we will maximise a lower bound of P(X). Let's define a distribution q(Z) that we know so that q(Z) approximates p(Z|X). Our goal is to minimise the KL divergence between q(Z) and p(Z|X).



Start: $KL[q(Z), p(Z|X)] = \sum q(Z)\, log \frac{q(Z)}{p(Z|X)}$

$$= -\sum q(Z)\, log \frac{p(Z|X)}{q(Z)} \quad (1)$$

Bayes Rule $P(Z|X) = \frac{p(X|Z).p(Z)}{p(X)}$

$$\xrightarrow{(1)} = -\sum q(Z)\, log \frac{\frac{p(X|Z).p(Z)}{p(X)}}{q(Z)}$$

$$\xrightarrow{p(X|Z) = \frac{p(X,Z)}{p(Z)}} = -\sum q(Z)\, log \frac{\frac{p(X,Z).p(Z)}{p(X).p(Z)}}{q(Z)}$$

$$= -\sum q(Z)\, log \frac{p(X,Z)}{q(Z)p(X)}$$

$$= -\sum q(Z)[log \frac{p(X,Z)}{q(Z)} + log \frac{1}{p(X)}]$$

$$= -\sum_z q(Z) \log \frac{p(X,Z)}{q(Z)} + \sum_z q(Z) \log p(X)$$

$$= -\sum_z q(Z) \log \frac{p(X,Z)}{q(Z)} + logp(X) \sum_z q(Z) \quad (2)$$

(2) $KL[q(Z), p(Z|X)] = -\sum_z q(Z) \log \frac{p(X,Z)}{q(Z)} + logp(X) \sum_z q(Z)$

constant, no Z

define as L

$\sum_z q(Z) = 1$

$$KL[q(Z), p(Z|X)] = -L + logp(X)$$

$$KL[q(Z), p(Z|X)] + L = logp(X)$$

≥ 0

$$\boxed{logp(X) \geq L}$$

Maximising L means maximising log p(X). Therefor we call L the variational lower bound for log p(X). If you maximise the lower bound of a function, you automatically maximise the function.

$$L = -\sum_z q(Z) \log \frac{p(X,Z)}{q(Z)}$$

$$\overset{\text{joint}}{=} -\sum_z q(Z) \log \frac{p(X|Z)p(Z)}{q(Z)}$$

$$= -\sum_Z q(Z) [\log p(X|Z) + \mathrm{q(Z)} \log \frac{p(Z)}{q(Z)}]$$

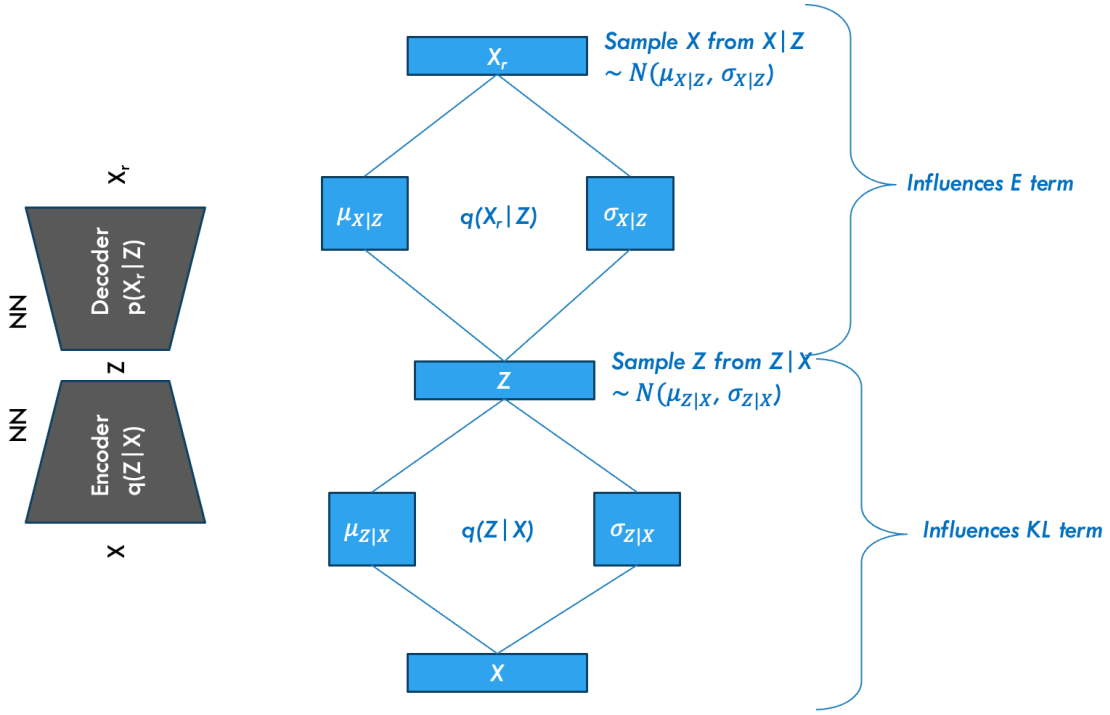
$E_{q(Z)} \log p(X|Z)$ -KL(q,p)

$$\boxed{L = E_{q(Z)} \log p(X|Z) - \mathrm{KL}\,(\mathrm{q(Z)}, \mathrm{p(Z)})]}$$

In the case of VAEs, training means maximising L, meaning make KL small and Eq(z) large. Instead of a deterministic function between Z and Xr, we assume a Gaussian relation.

VAEs at Work

This image illustrates the architecture and workflow of a Variational Autoencoder, focusing on the relationship between the encoder, decoder, and the probabilistic sampling process.



Encoder q(X|Z)

- The encoder maps an input X to a latent space Z
- Instead of directly outputting Z, the encoder predicts a mean μZ|X and standard deviation σZ|X for Z, assuming a Gaussian distribution for Z.
- The encoder aims to approximate q(Z|X) which influences the KL divergence term by ensuring the latent space distribution aligns with a prior p(Z)

Latent Space Sampling:

- The latent variable Z is sampled from the Gaussian distribution defined by the encoder parameters μZ|X and σZ|X using the reparameterisation trick. This trick separates the randomness ϵ from the deterministic parts μ and σ

Decoder P(X|Z):

- The decoder models the likelihood of reconstructing the input X given Z
- The reconstruction quality influences the E-term (reconstruction loss), which compares the reconstructed output Xr to the original input X.

Objective Function:

- The total loss combines:
  - The KL divergence term ensures the latent space distribution q(Z|X) is close to a predefined prior P(Z|X)
  - The Reconstruction loss (E-term) measures how accurately the decoder reconstructs the input.

Workflow:

1. Data X is passed into the Encoder produces μZ|X and standard deviation σZ|X for the latent distribution
2. Z is sampled from the Gaussian distribution defined by μZ|X and σZ|X
3. The sampled Z is fed into the Decoder to generate Xr
4. Loss Calculation:
    - The reconstruction loss evaluates how close Xr is to X
    - The KL divergence ensures regularisation in the latent space
5. Optimisation:
    - The combined loss is minimised during training to refine both the Encoder and Decoder.

VAEs integrate probabilistic sampling with neural networks to learn a structured latent representation while balancing reconstruction quality and latent space regularity.